Checkpoint 02 Report — Team Progress & Comparison with Checkpoint 01

CPU Priority Scheduling Simulation

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1) Executive summary

This document summarizes our progress from **Checkpoint 01 → Checkpoint 02** for the Priority Scheduling OS project.

Compared to Checkpoint 01, Checkpoint 02 delivers:

- **Full correctness coverage**: unit + integration + GUI tests are implemented and executed via `ctest` (3/3 suites).
- **Major performance improvements**: Preemptive priority scheduling becomes **event-based** and runs efficiently up to **1,000,000 processes**.
- **Scalability in GUI**: tables and Gantt visualization are adapted for very large datasets (including horizontal scroll for dense segments).

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2) Comparison with Checkpoint 01

2.1 Checkpoint 01 (baseline)

Checkpoint 01 reported the following baseline status (extracted from `CheckPoint/Checkpoint01_10/Checkpoint01_10_Report.pdf`):

Functional scope (demo-ready):

- Load valid CSV; run **Non-preemptive** and **Preemptive** variants (NP-only, P-only, or both).
- Display CT/TAT/WT/RT; show averages and NP/P comparison.
- Render **Gantt chart** (including IDLE segments) and export CSV results.

Testing:

- `ctest`: **3/3 suites pass**.
- **63 test cases, 3408 assertions** total.
- `unit_tests`: 53 cases / 3311 assertions (Pass)
- `integration_tests`: 1 case / 1 assertion (Pass, **stub**)
- `gui_tests`: 9 cases / 96 assertions (Pass)

Benchmark (reference, on developer's i5-11400F, Release O2):

- Non-preemptive:
- 100,000: **29 ms**
- 1,000,000: **0.86 s**
- 2,000,000: **2.00 s**
- Preemptive:
- 100,000: **311.35 s**
- 1,000,000 / 2,000,000: **not measured** (P estimated to take hours)
- Reported complexity concern: **O(n × total burst)** for Preemptive.

GUI limitations:

- GUI results table limited to **10,000 rows**.
- For datasets around **1,000,000 processes**, the GUI could not show everything; **Gantt chart was compressed**.

Main remaining issues / planned work (Checkpoint 01 → 02):

- Convert Preemptive implementation to **event-based + heap**.
- Add **benchmark runner** (CSV + reproducible logs).
- Improve GUI scalability (virtual table + Gantt adapted to viewport).
- Strengthen integration test to verify CSV → scheduler → output end-to-end.

2.2 Checkpoint 02 (current state)

Checkpoint 02 is fully implemented in the repository and validated by the automated test suite.

Testing (now fully integrated):

- `ctest` : **3/3 suites pass**
- Total: **99 test cases / 112,740 assertions**
- `unit_tests` : 79 / 112,207
- `integration_tests` : 10 / 318
- `gui_tests` : 10 / 215

Performance benchmarks (release configuration):

- **Non-preemptive (NP):**
- 100,000: **13.7 ms**
- 1,000,000: **242 ms**
- 2,000,000: **577 ms**
- **Preemptive (P):**
- 100,000: **27 ms**
- 500,000: **242 ms**
- 1,000,000: **572 ms**

Benchmark methodology:

- NP: measured as mean/median over 5 runs.
- P: measured for representative input sizes (single-run in our runner) after moving to event-based simulation.

GUI scalability & Gantt improvements (1,000,000-row file):

- `loadCsv` : **272 ms**
- Run NP only: **251 ms**
- Run both NP+P: **835 ms**
- Peak RSS: **247 MiB**
- Gantt rendering draws only visible segments and supports **horizontal scroll** when the dataset contains many segments.

2.3 Summary table — What improved?

Area	Checkpoint 01	Checkpoint 02
Testing coverage	63 cases / 3408 assertions, integration end-to-end mostly missing (stub)	99 cases / 112,740 assertions, integration verifies CSV → scheduler → export
Preemptive performance	100k ~311.35 s; complexity concern O(n × total burst)	Event-based simulation: 1,000,000 ~572 ms
GUI scalability	Table limited to ~10,000 rows; 1M view not feasible; Gantt compressed	Virtual `QTableView` + viewport-based Gantt; 1M rows viewable; horizontal scroll for dense segments
Benchmark reproducibility	Benchmark data measured manually / temporary tools	`benchmark_runner` + `benchmark_np` + report logs integrated
CSV error handling	Error reporting stopped early / limited visibility (as reported)	Error collection aggregates invalid lines up to a controlled limit

3) Team progress (Week-by-week milestones)

Week	Workstream	Deliverables	Status
1	Project foundation	CMake, conventions, frozen API contract	Done
2	CSV module	`CsvHandler`, CSV validation + unit tests (C01–C25)	Done
3	Non-preemptive Priority	`NonPreemptiveScheduler` + tests (N01–N18)	Done
4	Preemptive Priority	`PreemptiveScheduler` + tests (P01–P18)	Done
5	GUI skeleton + integration	`MainWindow`, `GanttChart`, GUI wiring + tests (U01–U13)	Done
6	GUI scalability	Virtual `QTableView` model + horizontal-scroll Gantt	Done (merge PR #40)
7	Benchmark + report data	`benchmark_runner`, `benchmark_np`, `generate_perf`	Done
8	Verification + final docs	`ctest` suites, final report + demo reference	Code ready; report finalized

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4) Member responsibility

Responsibility is aligned with Checkpoint 01’s member list, and updated to reflect the actual deliverables merged in the final implementation.

#	Member	MSSV	Responsibility
1	Võ Nguyễn Minh Hưng (Leader)	087207010338	Architecture, integration, benchmark orchestration, final QA/report
2	Huỳnh Tuấn Kiệt	066207016352	Non-preemptive scheduler implementation + unit tests
3	Nguyễn Trường Vĩ	095207004773	Preemptive scheduler implementation + unit tests
4	Lê Đình Minh Hưng	045207004732	Qt GUI implementation (MainWindow/ Gantt/virtual table) + GUI tests
5	Nguyễn Kim Hoàng Lan	068307000700	CSV I/O, validation rules, integration-level IO testing

Concrete deliverables (merged into the codebase):

- `Kiệt NP`: `src/core/NonPreemptiveScheduler.{h,cpp}`, `tests/unit/non-preemptive/*`
- `Vĩ P`: `src/core/PreemptiveScheduler.{h,cpp}`, `tests/unit/preemptive/*`
- `Minh Hưng GUI`: `src/ui/MainWindow.{h,cpp}`, `src/ui/GanttChart.{h,cpp}`, `src/ui/ProcessTableModel.{h,cpp}`, `tests/gui/gui_test.cpp`
- `Lan CSV`: `src/io/CsvHandler.{h,cpp}`, `tests/unit/csv/csv_handler_test.cpp`
- `Hưng (Leader)`: integration glue, benchmark tools, final report documentation